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Master's in Formal Methods in Computer Science

Master's Thesis

Type Theory and Categorical Models
A Unified Approach

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Abstract

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Resumen

Keywords:

0.1 Untyped λ-Calculus

In order to get acquainted with the λ-calculus, let us develop a simple example to familiarize ourselves before we begin with a more formal approach to this discipline. Consider the function $f(x) = x + 1$, the most straightforward way to express this using λ-calculus would be $(\lambda x.x + 1)$, where the lambda denotes that x is being captured and used as a parameter to perform some computation[1]. Evaluating $f(2)$ using this newly created lambda term would look like this: $(\lambda x.x + 1)(2) \rightsquigarrow_\beta (2 + 1) \rightsquigarrow_\beta 3$. This process of reducing a λ-expression is referred to as β-reduction, it will be covered formally later in the chapter.

0.1.0.1 The Language of λ-Calculus

Definition 0.1.1. The set of all lambda terms Λ is defined inductively as follows:

- (Variable) If $x \in \mathbf{V}$, then $x \in \Lambda$.
- (Abstraction) If $x \in \mathbf{V}$ and $M \in \Lambda$, then $(\lambda x.M) \in \Lambda$.
- (Application) If $M, N \in \Lambda$, then $(MN) \in \Lambda$.

Where $\mathbf{V} = \{x, y, z, ...\}$ represents a countably infinite set of variable names.

The key takeaway of this definition is that abstraction and application together, when combined with β-reduction, enable computation—thus encapsulating the meaning of function in a way that is abstract yet useful.

Notation 1. We use uppercase letters, e.g. M, N, L, P, Q to denote elements of Λ and lowercase e.g. x, y, z to denote elements of $\mathbf{V}$.

Since we are dealing with a formal language, it is in our benefit to introduce a few other objects with the aim of defining a grammar to generate it:

- An alphabet $\Sigma = \{\lambda, ., (,), \ldots\}$, is a finite set of symbols
- A string is a finite sequence of elements from Σ, the empty string is denoted by ε
- Σ^* denotes the set of all finite strings over Σ, $\varepsilon \in \Sigma^*$
- A language L over an alphabet Σ is a subset of Σ^*

Our aim now, to generate the set Λ, to do this, we will make use of a grammar. When dealing with languages, notation and similar concepts, Backus-Naur Form, often shortened to BNF, is one of the more complete alternatives:

- Nonterminals are enclosed in angle brackets (e.g. `<expr>`)

[1] this is an example, $(\lambda x.x + 1)$ is not a valid lambda term, see Definition 0.1.2.

- Terminals are written literally (e.g. "λ", ".", x)
- Productions define how nonterminals expand, written as ::=
- The vertical bar | denotes available expansions

Thus, the language for natural numbers in decimal notation is represented using:

```
<digit> ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"
<number> ::= <digit> | <digit> <number>
```

Notation 2. Throughout the document BNF will be used to define language specifics. In other words, it will be used when details matter. The times when notational rigor is beyond the point, abstract syntax will be used to convey more general concepts that do not require a standard notation.

Definition 0.1.2. Taking advantage of BNF notation, an alternative definition for Λ would be:

```
<term> ::= <variable>
         | "λ" <variable> "." <term>
         | <term> <term>
         | "(" <term> ")"
<variable> ::= ∈ V
```

where $\Sigma = \{\lambda, ., (,)\} \cup \mathbf{V}$ and $\Lambda \subset \Sigma^*$. Through use of inductive types types[2], this grammar can be implemented in Lean4:

Remark 1. From the definition above we extract that $\lambda x.x + 1 \notin \Lambda$, given this circumstance, the need to find an accurate representation for numbers within this definition arises. See Definition **??**

Example 0.1.1. Some examples of valid λ terms generated using the grammar in 0.1.2: y, $(\lambda x.(xx))$, $(\lambda x.(\lambda y.x))$, $(((\lambda x.(xy))(\lambda y.y))z)$. The Lean4 counterpart for the second term is:

As a convention to avoid notational cluttering, the outermost parenthesis can be omitted e.g. $\lambda x.x$ is read as $(\lambda x.x)$. Also, application is left-associative and binds tighter than abstraction, so MNP is parsed as $(MN)P$ and $\lambda x.MN$ means $\lambda x.(MN)$, not $(\lambda x.M)N$. Likewise, abstraction is right-associative e.g. $\lambda x.\lambda y.M$ is $\lambda x.(\lambda y.M)$ and binds more weakly than application. Haskell Curry introduced what we now referr to as currying, its main purpose is to avoid unnecessary lambdas. Below is a summary of common notational conventions used throughout the text:

[2] Inductive Data Types are built from sums (either this or that) of products (this and that). The main point is that they allow self-reference and support pattern matching.

- $\lambda x.x \iff (\lambda x.x)$
- $MNP \iff (MN)P$
- $\lambda x.MN \iff \lambda x.(MN)$
- $\lambda x.\lambda y.f(x,y) \iff \lambda xy.f(x,y)$

All these are just to keep notation straight-forward, the formal definition does not leave precedence under-specified thanks to parenthesis. Those interested in the implementation need not worry, as Lean4 enforces an explicit order on constructors.

0.1.1 Equivalence of terms

Now that the λ-calculus language has been introduced, it is time we move on to the mechanics of computation, manifested through β-equivalence. α and η equivalence will also be introduced. These two are more difficult to grasp. Informally, it could be said that α-equivalence is equivalence programs up to renaming of local variables. In this same scenario, η-equivalence would singnify equivalence of programs up to wrapping or unwrapping them with an identity function.

0.1.1.1 Subterms

From the ground up, we begin define the concept of subterm, since the definition of the language is self-referential and terms might contain other terms.

Definition 0.1.3. Given $T \in \Lambda$, $\mathbf{Sub} : \Lambda \to \mathscr{P}(\mathbf{V})$ maps a term to the set of it's subterms.

$$\begin{aligned}
&\mathbf{Sub}(x) = \{x\} \\
&\mathbf{Sub}(MN) = \mathbf{Sub}(M) \cup \mathbf{Sub}(N) \cup \{MN\} \\
&\mathbf{Sub}(\lambda x.M) = \mathbf{Sub}(M) \cup \{\lambda x.M\}
\end{aligned}$$

Where $x \in \mathbf{V}$ and $M, N \in \Lambda$. In Lean4:

Remark 2. Let $\preceq$ be the subterm relation on λ-terms, defined by

$$M \preceq N \iff M \in \mathbf{Sub}(N)$$

$\preceq$ is a partial order relation. This statement is follows intuitively from the definition, it can be proven using induction on the derivation tree.

Definition 0.1.4. A term $S \in \mathbf{Sub}(M)$, $M \in \Lambda$ is said to be proper if $S \neq M$. $\mathbf{Sub}(M) \setminus \{M\}$ would be the set of proper subterms.

Definition 0.1.5. Let $M = \lambda x.N$ be a λ-term:

- The variable x is the binding variable of the abstraction $\lambda x.N$.
- An occurrence of a variable x in M is bound if $x \in \mathbf{Sub}(\lambda x.N)$.
- Free variables are those that are not bound.

Example 0.1.2. Take $M \equiv \lambda x.\, x\, y$, in this example, the x in $\lambda x.$ is binding, y is free, and x is bound. This means that y is used literally, and that x, as denoted by $\lambda x.$ is abstracted and thus subject to being replaced within the context of said abtraction. A subexample might make this explicit: $(\lambda x.\, x\, y)\, M \rightsquigarrow_\beta M\, y$. The y remained while the x was used as a placeholder for M.

Note 1. This taxonomy is key to understanding concepts such as α-equivalence intuitively.

0.1.1.2 Free Variables

The set of Free Variables **FV**, those that will not get replaced during the process of reduction, is computed using:

Definition 0.1.6. Given $T \in \Lambda$, $\mathbf{FV} : \Lambda \to \mathscr{P}(\mathbf{V})$ outputs the set of free variables for T:

$$\begin{aligned}
&\mathbf{FV}(x) = \{x\} \\
&\mathbf{FV}(MN) = \mathbf{FV}(M) \cup \mathbf{FV}(N) \\
&\mathbf{FV}(\lambda x.M) = \mathbf{FV}(M) \setminus \{x\}
\end{aligned}$$

Where $x \in \mathbf{V}$ and $M, N \in \Lambda$, and $\mathscr{P}$ denotes the power set.

Example 0.1.3. Computing the free variables of $\lambda x.\lambda y.yxz$. The term well-formed, so we proceed:

$$\begin{aligned}
\mathbf{FV}(\lambda x.\lambda y.yxz) &= \mathbf{FV}(\lambda y.yxz) \setminus \{x\} \\
&= \mathbf{FV}(yxz) \setminus \{x, y\} \\
&= \{x, y, z\} \setminus \{x, y\} \\
&= \{z\}
\end{aligned}$$

Of course, z is the only free variable in the expression as it is the only variable that is not captured by some λ-abstraction.

Definition 0.1.7. A term $M \in \Lambda$ is closed if and only if $\mathbf{FV}(M) = \emptyset$, closed lambda terms are often referred to as combinators. Λ^0 is the set of all closed lambda terms.

Note 2. They are called combinators since having no free variables means that all variables are bound, and thus, a closed term just combines bound variables.

0.1.1.3 Variable Substitution

We approach the definition for α-equivalence to this end we have to introduce what it means to substitute a variable.

Definition 0.1.8. Capture-Avoiding Substitution Rules Let $M, N \in \Lambda$ and $x, y, z \in \mathbf{V}$. The substitution $M[x := N]$ is defined inductively as:

$$
\begin{aligned}
x[x := N] &= N \\
y[x := N] &= y && y \neq x \\
(M_1\, M_2)[x := N] &= (M_1[x := N])\,(M_2[x := N]) \\
(\lambda x.M)[x := N] &= \lambda x.M \\
(\lambda y.M)[x := N] &= \lambda y.(M[x := N]) && y \neq x, y \notin \mathbf{FV}(N) \\
(\lambda y.M)[x := N] &= \lambda z.(M[y := z][x := N]) && y \neq x, y \in \mathbf{FV}(N)
\end{aligned}
$$

where $z \notin \mathbf{FV}(M) \cup \mathbf{FV}(N)$.

Hopefully the meaning of the first three rules is clear, they ancapsulate renaming in the context of variables and applications. The subsequent rules are about abstractions, rule four states that renaming a binding variable has no effect, renaming only modifies a term whenever the renamed variable is different from the binding variable. Rule five is the general case, that is, when the substitution just moves in together with the abstracted term. Then there is the last case, when the binding variable is a bound occurrence in the term we are substituting, which poses a problem, to see why, let us work through an example:

Example 0.1.4. If we mistakenly substitute x by $(\lambda w.\, w\, y)$ in $(\lambda y.\, y\, x)$ using rule five, we get:

$$(\lambda y.\, y x)[x := (\lambda w.\, w\, y)] = (\lambda y.\, y\, (\lambda w.\, w\, y))$$

This is problematic since the free variable y became bound after the rename. Free and bound variables should mantain their freedom or their boundedness after the rename.

Note 3. This is essentially why it is called Capture Avoiding Substitution and not simply Substitution.

After introducing rule six, we rename the binding variable to avoid the name clash:

$$(\lambda y.\, y x)[x := (\lambda w.\, w\, y)] = (\lambda z.\, z\, (\lambda w.\, w\, y))$$

The purpuse of the binding variable is to serve as a placeholder, therefore renaming does not affect the integrity of the expression, the variable gets replaced anyway when we perform a reduction.

In the previous example a bound variable is renamed to avoid a naming collision and avoid binding a previously free variable, α-equivalence is the generalization of this technique, it defines equivalence up to renaming of binding variables.

One notable result regarding substitution is the substitution lemma:

Lemma 0.1.1. Let M, N, P be lambda terms and x, y be variables. Then:

$$M[x := N][y := P] = \begin{cases} M[x := N[y := P]], & \text{if } x = y, \\ M[y := P][x := N[y := P]], & \text{if } x \neq y \text{ and } y \notin \mathrm{FV}(N). \end{cases}$$

Proof. structural induction on M. □

Notation 3. In many of the upcoming definitions inference rules are used, here is a simple introduction for those unfamiliar with the concept:

$$\frac{P_1 \quad P_2 \quad \cdots \quad P_n}{C}\ \textsc{Name}$$

where P_i are judgments assumed to hold, C is the judgment inferred from the premises and NAME is an optional identification label.

0.1.1.4 α-equivalence

Definition 0.1.9. Two lambda terms $M, N \in \Lambda$ are α-equivalent, written $M =_\alpha N$, if they are structurally identical except for the names of bound variables. Formally:

$$\frac{}{\lambda x.L =_\alpha \lambda y.L[x := y]}\ \textsc{Alpha} \qquad \frac{L =_\alpha L'}{\lambda x.L =_\alpha \lambda x.L'}\ \textsc{Abs}$$

$$\frac{L_1 =_\alpha L_1'}{L_1\, L_2 =_\alpha L_1'\, L_2}\ \textsc{AppL} \qquad \frac{L_2 =_\alpha L_2'}{L_1\, L_2 =_\alpha L_1\, L_2'}\ \textsc{AppR}$$

α-equivalence is an equivalence relation:

(Reflexivity)	$M =_\alpha M$
(Transitivity)	$M =_\alpha N$ and $N =_\alpha P \Rightarrow M =_\alpha P$
(Symetry)	$M =_\alpha N \Rightarrow N =_\alpha M$

Example 0.1.5. A simple equivalence class and two samples of not α-equivalent pairs.

$$\lambda x.\lambda y.\, y =_\alpha \lambda a.\lambda b.\, b =_\alpha \lambda z.\lambda x.\, x$$

$$\lambda x.\lambda y.\, x \neq_\alpha \lambda x.\lambda y.\, y \quad \lambda x.\, x\, y \neq_\alpha \lambda x.\, x\, z$$

Remember terms are α-equivalent only when bounded names are properly substituted.

Remark 3. α-equivalence comes naturally once substitution is defined, since the names of bound variables are just placeholders that will eventually hold an argument, thus getting replaced.

Note 4. Variable naming is a nightmare. De Bruijn indexing is a way of methodically assigning numbers to variable names, indicating how many binders away a variables binder is. This system eliminates the need for α equivalence to exist.

$$\lambda x.\, x \Rightarrow \lambda.0 \quad \lambda x.\lambda y.\, x \Rightarrow \lambda.\lambda.1 \quad \lambda x.\lambda y.\, y \Rightarrow \lambda.\lambda.0$$

We are better at names than we are at numbers, which is why we will prefer using symbols over De Brujn indexing. Programs like automated theorem provers and compilers do use similar naming strategies to simplify their work.

0.1.1.5 β-equivalence

Definition 0.1.10. Single-step β-reduction $\rightsquigarrow_\beta$ is defined using the following inference rules:

$$\frac{}{(\lambda x.\, M)\, N \rightsquigarrow_\beta M[x := N]}\ \textsc{Beta} \qquad \frac{M \rightsquigarrow_\beta M'}{\lambda x.\, M \rightsquigarrow_\beta \lambda x.\, M'}\ \textsc{Abs}$$

$$\frac{M_1 \rightsquigarrow_\beta M_1'}{M_1\, M_2 \rightsquigarrow_\beta M_1'\, M_2}\ \textsc{AppL} \qquad \frac{M_2 \rightsquigarrow_\beta M_2'}{M_1\, M_2 \rightsquigarrow_\beta M_1\, M_2'}\ \textsc{AppR}$$

where $M, N \in \Lambda$.

Single β-reductions can be applied succesively, which induces the following definition

Definition 0.1.11. Multi-step β-reduction. We write $M \rightsquigarrow_\beta^* N$ if and only if there exists a sequence of one-step β-reductions starting from M and ending at N.[3]

$$M \equiv M_0 \rightsquigarrow_\beta M_1 \rightsquigarrow_\beta M_2 \rightsquigarrow_\beta \cdots \rightsquigarrow_\beta M_{n-1} \rightsquigarrow_\beta M_n \equiv N$$

That is, there exists an integer $n \geq 0$ and a sequence of terms $M_0, M_1, \ldots, M_n$ such that: $M_0 \equiv M$ and $M_n \equiv N$ for all i with $0 \leq i < n$, we have $M_i \rightsquigarrow_\beta M_{i+1}$.

Note 5. Multi-step $\rightsquigarrow_\beta^*$ is the reflexive transitive closure of $\rightsquigarrow_\beta$.

The process of succesively β-reducing a term leads raises the question of whether all terms can be reduced indefinitly. We know the answer to this, since for instance, variables in **V** cannot be reduced. The subsequent question is then, Do all all terms have some kind of normal form that cannot be reduced further? This is answered in Section **??**. To this end we introduce a formal definition of normal form that will allow us to gain insight into the normalization properties of the λ-calculus.

[3] The convention in the literature is to use $\rightarrow_\beta$ for β-reductions $\twoheadrightarrow_\beta$ for multi-step β-reductions. I prefer to use $\rightsquigarrow_\beta$ and $\rightsquigarrow_\beta^n$ since this allows me to use the superscript to note the number of steps in the process β-reduction e.g. $(\lambda f.\lambda x.\, x)\, M\, N \rightsquigarrow_\beta^2$ N. Instead of $\rightsquigarrow_\beta^n$ I use $\rightsquigarrow_\beta^*$ whenever the number of steps is undefined.

Definition 0.1.12. A λ-term M is in normal form if there is no term N such that $M \rightsquigarrow_\beta N$. Equivalently, M contains no subterm of the form $(\lambda x.P)\,Q$ that can be reduced by beta-reduction. In other words, M is irreducible under beta-reduction.

Definition 0.1.13. A reducible expression, or redex for short, is any subterm of the form $(\lambda x.P)\,Q$. Such a term can be reduced by a single beta-reduction step, resulting in $P[x := Q]$.

Definition 0.1.14. β-equivalence relation, equivalence of λ-terms through β-reductions.

$$M =_\beta N \iff M \rightsquigarrow_\beta^* N \vee N \rightsquigarrow_\beta^* M$$

Note 6. $=_\beta$ is the symetric closure of $\rightsquigarrow_\beta^*$.

We now explore a key aspect of the λ-calculus, in the jargon referred to as first-class citizenship, or high-order, a property of terms that restrains them from being exclusively classified as either function or argument. The example below sheds some light on the matter:

Example 0.1.6. High Order by example. Let us β-reduce the λ-term ABC and inspect the consequences:

$$\begin{aligned}
\overbrace{(\lambda f.\lambda y.f\,f\,y)}^{A}\overbrace{(\lambda x.x+1)}^{B}\overbrace{(2)}^{C} &\rightsquigarrow_\beta (\lambda y.(\lambda x.x+1)(\lambda x.x+1)\,y)(2) \\
&\rightsquigarrow_\beta (\lambda x.x+1)(\lambda x.x+1)(2) \\
&\rightsquigarrow_\beta^* (\lambda x.x+1)(3) \\
&\rightsquigarrow_\beta^* (4)
\end{aligned}$$

Taking a closer look, A, applies f twice to x, as a consequence, B is being applied twice to C. And so, very easily, we have implemented the $x+2$ function via a sequential application of $x+1$ to C.

Remark 4. As seen above, we can pass what we understand as conventional functions as arguments since they are treated alike. One can intuitively apreciate the computational expresivenes this brings with it, and how this syntactic-semantic homogeneity sets the λ-calculus apart from the classical set-theoretic approach to functions.

0.1.1.6 η-equivalence

The last and indeed the least important notion of equivalence. It refers to a program being equivalent to itself wrapped with an identity function:

Definition 0.1.15. Two lambda terms M and N are said to be η-equivalent, written $M =_\eta N$, if:

$$\frac{x \notin \mathbf{FV}(M)}{(\lambda x.M) =_\eta M}\ \textsc{Eta} \qquad \frac{M =_\eta M'}{\lambda x.M =_\eta \lambda x.M'}\ \textsc{Abs}$$

$$\frac{M_1 =_\eta M_1'}{M_1\, M_2 =_\eta M_1'\, M_2}\ \textsc{AppL} \qquad \frac{M_2 =_\eta M_2'}{M_1\, M_2 =_\eta M_1\, M_2'}\ \textsc{AppR}$$

The abstraction $\lambda x.(M\ x)$ performs the same operation as M, so if x is not used freely within M, the abstraction serves no purpose. This is shown in Eta, the rest of the inference rules propagate this equivalence throughout the whole term.

Note 7. Two terms are observationally equivalent if they cannot be distinguished by any context that can be papplied to them. η-equivalence can be understood as a form of weak extensionality, since it does not fully encode observational equivalence.

Add something about normalization properties and how they dont ensure termination. this is important for programming blah blah blah... This is why we introduce Types and Type Systems... This notion is similar to that of restricting the domain and codomain of a function.